Energy in the multimode cavity

$$\hat{H} = \frac{1}{2}\int_{cavity}\left[\varepsilon_0 \hat{E}_T \cdot \hat{E}_T + \mu_0^{-1}\hat{B}\cdot\hat{B}\right]dV$$

$$H = \sum_{k\lambda} \hbar\omega_k\left[\hat{a}^+_{k\lambda}\hat{a}_{k\lambda} + \frac{1}{2}\right]$$

$$\hat{H}\left|\{n_{k\lambda}\}\right\rangle = \sum_{k\lambda}^{\infty} \hbar\omega_k\left(n_{k\lambda} + \frac{1}{2}\right)\left|\{n_{k\lambda}\}\right\rangle$$ Eigenvalue equation

$$E_0 = \sum_{k\lambda}^{\infty} \frac{1}{2}\hbar\omega_k = \sum_{k}^{\infty}\hbar\omega_k$$ → Infinite energy of the ground state $[n_{k\lambda}=0\ \forall k,\lambda]$

$$E_{Rad} = \sum_{k\lambda}\hbar\omega_k n_{k\lambda}$$ → Radiative energy corresponding to sum of the energy of all photons

$$E_{TOT} = E_0 + E_{rad}$$ → Total EM energy in the quantization cavity

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Single mode states → Number States or Fock States {Plane travelling waves}

Fock States are the eigenvectors of the Hamiltonian for a single mode and form a complete set of states → Basic states for the quantum theory of light.
Ideal base for the blackbody distribution. They cannot be easily generated experim.
Let's drop $K\lambda$ (Defined)

$\hat{n}|n\rangle = n|n\rangle$ implies a defined photon number in the mode n

$(\Delta n)^2 = 0$ In fact $\langle n^2\rangle = \langle n|\hat{n}^2|n\rangle = n^2 \ \forall n$ $(\Delta n)^2 = \langle n^2\rangle - \langle n\rangle^2 = 0$

$\hat{H}|n\rangle = E_n|n\rangle$ → The energy is also defined $E_n = (n+1/2)\hbar\omega$ $[\omega = KC]$

$\hat{H} = \hbar\omega(\hat{X}^2+\hat{Y}^2) \Rightarrow (\hat{X}^2+\hat{Y}^2)|n\rangle = (n+1/2)|n\rangle$ $\left\{\hat{X} = 1/2(\hat{a}^+ + \hat{a}) \quad \hat{Y} = \frac{i}{2}(\hat{a}^+ - \hat{a})\right\}$ ←

The expectation Value for the quadrature operators are:

$\langle n|\hat{X}|n\rangle = 1/2\langle n|\hat{a}^+ + \hat{a}|n\rangle = 1/2\left[\langle n|\hat{a}^+|n\rangle + \langle n|\hat{a}|n\rangle\right] = 0$ $\langle n|\hat{Y}|n\rangle = 0$

$\langle n|\hat{X}^2+\hat{Y}^2|n\rangle = n + 1/2$ → $\langle n|\hat{X}^2|n\rangle = 1/2(n+1/2)$ % $(\Delta X)^2 = \langle \hat{X}^2\rangle - \langle\hat{X}\rangle^2 = 1/2(n+1/2)$

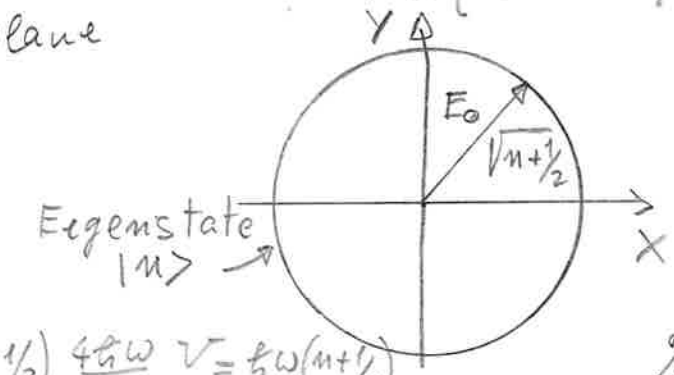
→ $\langle n|\hat{Y}^2|n\rangle = 1/2(n+1/2)$ $(\Delta Y)^2 = \langle\hat{Y}^2\rangle - \langle\hat{Y}\rangle^2 = 1/2(n+1/2)$

Uncertainty: $(\Delta X)^2(\Delta Y)^2 = (1/2\, n + 1/4)^2 \Rightarrow [n=0]$ $(\Delta X)^2(\Delta Y)^2 = 1/16$ min

Electric field in normalized units: $2\sqrt{\hbar\omega/2\varepsilon_0 V}$

$\hat{E}_T = \hat{E}_T^+(\chi) + \hat{E}_T^-(\chi) = 1/2\left[\hat{a}\exp(-i\chi) + \hat{a}^+\exp(i\chi)\right]$ $\chi = \omega t - Kz - \pi/2$ $\left\{\begin{array}{l}\hat{a} = \hat{X} + i\hat{Y} \\ \hat{a} = \hat{X} - i\hat{Y}\end{array}\right.$

$\hat{E}_T = \hat{X}\cos\chi + \hat{Y}\sin\chi$ The electric field represents a plane wave propagating along z



Expectation Value of $\hat{E}_T$

$S = \langle n|\hat{E}_T(\chi)|n\rangle = 0$ → Signal in a Homodyne experiment

$N^2 = \langle n|\hat{E}_T(\chi)^2|n\rangle = 1/2(n+1/2) = E_{RMS}^2 = E_0^2/2$

$E_0 = \sqrt{n+1/2}$ in units of $\underset{\uparrow}{Q}\sqrt{\hbar\omega/2\varepsilon_0 V}$ Magnetic field Energy $= \varepsilon_0 E_0^2 V = \frac{\varepsilon_0}{2}(n+1/2)\frac{4\hbar\omega}{2\varepsilon_0 V}V = \hbar\omega(n+1/2)$ %

$$S = \langle n|\hat{E}_T(x)|n\rangle = \langle n|\hat{x}|n\rangle \cos x + \langle n|\hat{y}|n\rangle \sin x = 0$$

$$M^2 = \langle n|\hat{E}_T^2(x)|n\rangle = \ ? \qquad \hat{E}_T^2 = \tfrac{1}{4}\left[\hat{a}\hat{a}\cdot\exp(-2ix) + \underbrace{\hat{a}\hat{a}^+ + \hat{a}^+\hat{a}} + \hat{a}^+\hat{a}^+\exp(2ix)\right]$$

$$\langle n|\hat{a}\hat{a}|n\rangle = \sqrt{n}\,\langle n|a|n-1\rangle = \sqrt{n}\sqrt{n-1}\,\langle n|n-2\rangle = 0$$

$$\langle n|\hat{a}^+\hat{a}^+|n\rangle = \sqrt{n+1}\sqrt{n+2}\,\langle n|n+2\rangle = 0$$

$$\langle n|\underbrace{\hat{a}\hat{a}^+ + \hat{a}^+\hat{a}}_{\frac{2}{\hbar\omega}\hat{H}}|n\rangle = \frac{2}{\hbar\omega}\langle n|\hat{H}|n\rangle = \frac{2}{\hbar\omega}(n+1/2)\hbar\omega = 2(n+1/2)$$

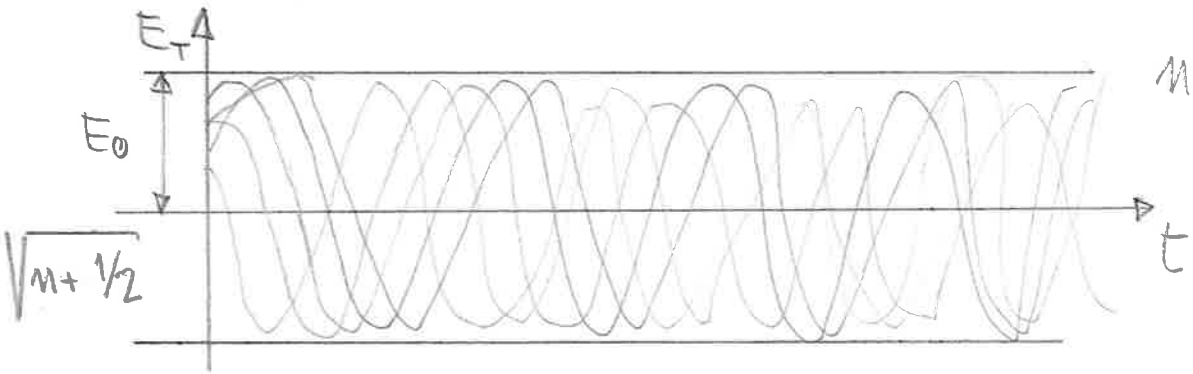
$$M^2 = \langle n|\hat{E}_T^2|n\rangle = \tfrac{1}{4}\left[2(n+1/2)\right] = \tfrac{1}{2}(n+1/2) = E_{T\,eff}^2 = E_0^2/2$$

$$E_{T\,eff}^2 = M^2 = \tfrac{1}{2}(n+1/2) \Rightarrow \text{Amplitude of } E_T:\ E_0 = \sqrt{n+1/2} \qquad \left[E_{T\,eff} = E_0/\sqrt{2}\right]$$

$$E_{T\,eff} = \frac{1}{\sqrt{2}}\sqrt{n+1/2}$$

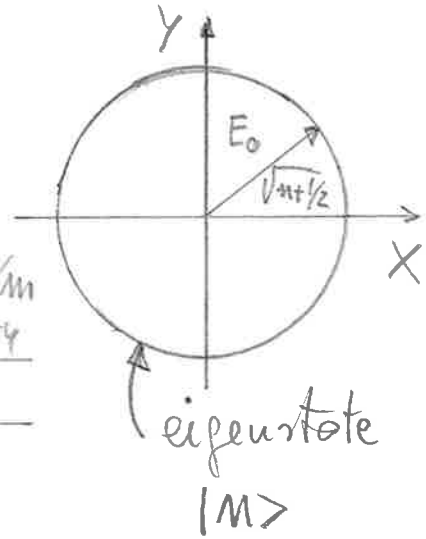
Magnetic Field

$$\text{Energy} = \varepsilon_0 E_{eff}^2 V = \varepsilon_0 \tfrac{1}{2} E_0^2 \left[2\sqrt{\hbar\omega/2\varepsilon_0 V}\right]^2 V = \frac{\varepsilon_0}{2}(n+1/2)\frac{4\hbar\omega}{2\varepsilon_0 V} V = (n+1/2)\hbar\omega$$



$n \gg 1$ $\quad 2\sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}} \simeq 10^{-2}\,V/m$

For a gas laser cavity

Phase unknown

$$E_0 = \sqrt{n+1/2}$$